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Combination (selection)

→ It is a selection of objects in which order is not important. AB & BA are different permutation but are same combination.

The number of combination of n -different objects taken r at a time is denoted by $C(n, r)$ or nC_r or $\binom{n}{r}$ or C_r^n .

Theorem:

The number of combinations of n -different objects taken r at a time is given by $C(n, r)$ or ${}^nC_r = \frac{n!}{(n-r)! r!}$.

Proof: Here,

$C(n, r)$ = no. of selection (combination) of n -different objects taken r at a time.

$P(n, r)$ = no. of permutation of n -different objects taken r at a time.

$r!$ = no. of arrangements of selected objects.

We know,

Selection $\times$ arrangements of selected objects = permutation,

$$\Rightarrow C(n, r) \times r! = P(n, r)$$

$$\Rightarrow C(n, r) = \frac{n!}{(n-r)!} \times \frac{1}{r!}$$

$$\therefore C(n, r) = \frac{n!}{(n-r)! r!}$$

$$\#$$

• Corollary :

① When $r=0$,

$$c(n,0) = \frac{n!}{(n-0)! 0!} = \frac{n!}{n! \times 1} = 1$$

② When $r=1$,

$$c(n,1) = \frac{n!}{(n-1)! 1!} = \frac{n(n-1)!}{(n-1)!} = n$$

③ When $r=n$

$$c(n,n) = \frac{n!}{(n-n)! n!} = \frac{n!}{0! n!} = 1$$

• Properties of combination

① The number of combination of n -different object taken r at a time and $n-r$ at a time are equal.

$$\boxed{\text{i.e. } c(n,r) = c(n,n-r)}$$

i.e. no. of selection = no. of elimination.

Proof

$$\text{R.H.S} = c(n,n-r)$$

$$= \frac{n!}{(n-(n-r))! (n-r)!}$$

$$= \frac{n!}{r! (n-r)!}$$

$$= c(n,r)$$

$$= c(n,r)$$

$$\text{Proved}$$

② If $c(n,r) = c(n,r')$ then either $r=r'$
or $r+r'=n$

Proof

$$\text{Given: } C(n, r) = C(n, r')$$

$$\Rightarrow r = r'$$

Now,

$$C(n, r) = C(n, r')$$

$$\Rightarrow C(n, r) = C(n, n-r') \quad [\because C(n, r') = C(n, n-r')]$$

$$\Rightarrow r = n-r'$$

$$\Rightarrow r + r' = n$$

$$\textcircled{3} \quad C(n, r) + C(n, r-1) = C(n+1, r) \quad [\text{Pascal's law}]$$

proof

$$\text{L.H.S} = C(n, r) + C(n, r-1)$$

$$= \frac{n!}{(n-r)! r!} + \frac{n!}{(n-r+1)! (r-1)!}$$

$$= \frac{n!}{(n-r)! r(r-1)!} + \frac{n!}{(n-r+1)! (r-1)!}$$

$$= \frac{n!}{(n-r)! r(r-1)!} + \frac{n!}{(n-r+1)(n-r)! (r-1)!}$$

$$= \frac{n!}{(n-r)! (r-1)!} \left[\frac{1}{r} + \frac{1}{n-r+1} \right]$$

$$= \frac{n!}{(n-r)! (r-1)!} \left[\frac{n-r+1+r}{r(n-r+1)} \right]$$

$$= \frac{n!}{(n-r)! (r-1)!} \left[\frac{(n+1)}{r(n-r+1)} \right]$$

$$= \frac{(n+1)!}{(n-r+1)! r!}$$

$$= C(n+1, r)$$

$$\textcircled{4} \quad \frac{C(n, r)}{C(n-1, r-1)} = \frac{n}{r}$$

Note:

① Total no. of handshakes among n -different people is

$$\begin{aligned} & c(n, 2) \text{ or } {}^nC_2 \\ &= \frac{n!}{(n-2)! \cdot 2!} \\ &= \frac{n(n-1)(\cancel{n-2}!) }{(\cancel{n-2}!) \cdot 2!} \\ &= \frac{n(n-1)}{2!} = \frac{n(n-1)}{2} \end{aligned}$$

② Total no. of diagonals of a polygon having n -sides

$$\begin{aligned} &= c(n, 2) - n \\ &= \frac{n!}{(n-2)! \cdot 2!} - n \\ &= \frac{n(n-1)(\cancel{n-2}!) }{(\cancel{n-2}!) \cdot 2!} - n \\ &= n \left[\frac{n-1}{2} - 1 \right] \\ &= n \left[\frac{n-1-2}{2} \right] \\ &= \frac{n(n-3)}{2} = \frac{n^2 - 3n}{2} \end{aligned}$$

Exercise 1.3

Part B

1.a) If $c(n, 10) = c(n, 12)$ find ' n ' and hence $c(n, 6)$

↳ Soln,

$$\begin{array}{l} r = 10, r' = 12 \\ r + r' = 10 + 12 = 22 = n \end{array} \left[\begin{array}{l} \text{If } c(n, r) = c(n, r') \text{ then} \\ r + r' = n \text{ or } r = r' \end{array} \right]$$

$$\therefore n = 22$$

Now,

$$\begin{aligned} C(n, 6) &= C(22, 6) \\ &= \frac{22!}{(22-6)! 6!} \\ &= \frac{22!}{16! 6!} \\ &= 74613 \end{aligned}$$

d) If $C(9, 2r) = C(9, 3r-1)$, find the value of r .

→ Soln, If $C(n, r) = C(n, r')$ then.

We have two condition,

$$r = r'$$

$$\text{or, } r + r' = n$$

$$\Rightarrow 2r = 3r - 1$$

$$\Rightarrow 2r + 3r - 1 = 9$$

$$\Rightarrow 3r - 2r = 1$$

$$\Rightarrow 5r = 10$$

$$\Rightarrow r = 1$$

$$\Rightarrow r = 2$$

$$\therefore r = 1, 2$$

Note: r should be Whole number less than or equal to n .

e) Prove that:

$${}^nC_r + 2 {}^nC_{r-1} + {}^nC_{r-2} = {}^{n+2}C_r$$

→ L.H.S,

$$\begin{aligned} & {}^nC_r + 2 \cdot {}^nC_{r-1} + {}^nC_{r-2} \\ &= C(n, r) + 2(C(n, r-1)) + C(n, r-2) \\ &= C(n, r) + C(n, r-1) + C(n, r-1) + C(n, r-2) \\ &= C(n+1, r) + C(n+1, r-1) \\ &= C(n+2, r) \quad (\text{using pascal's law}) \\ &= {}^{n+2}C_r \\ &= \text{R.H.S. proved} \end{aligned}$$

b) If $C(n, 8) = C(n, 6)$, find $C(n, 2)$

If $C(n, 8) = C(n, 6)$ then,

We have, ~~two conditions~~

or,

$$r + r' = n$$

$$\Rightarrow 8 + 6 = n$$

$$\Rightarrow n = 14$$

c) If $C(n, 30) = C(n, 4)$, find the value of $C(n, 30) + C(n, 4)$

$\rightarrow$ If $C(n, 30) = C(n, 4)$ then,

We have,

$$r + r' = n$$

$$\Rightarrow 30 + 4 = n$$

$$\therefore n = 34$$

Now,

$$C(34, 30) = \frac{34!}{(34-30)! \times 30!} = 46376$$

$$C(34, 4) = \frac{34!}{(34-4)! \times 4!} = 46376$$

~~or~~ Then,

$$C(34, 30) + C(34, 4) = 46376 + 46376 \\ = 92752 \text{ ways} \quad \neq!$$

2.

a) In an advertisement of 3 workers in a factory, if 10 applicants apply, how many ways can the selection be made?

$\rightarrow$ Soln,

Total applicant $(n) = 10$

Selected applicant $(r) = 3$ (To select)

Now,

$$C(n, r) = ?$$

We know,

$$\begin{aligned} C(10,3) &= \frac{10!}{(10-3)! \cdot 3!} \\ &= \frac{3628800}{7! \times 3!} \\ &= 120 \text{ ways} \end{aligned}$$

b) In a class of 25 girls and 20 boys, a boy and a girl are to be chosen for debate competition. In how many ways can the selection be made.

→ Soln,

$$\text{Girls} = 25$$

$$\text{Boys} = 20$$

We have to select a boy & a girl.

Girls [25]	Boys [20]	Selection
1	1	$C(25,1) \times C(20,1)$
		$= 500$

∴ no. of selection = 500

c) A person has 12 friends of whom 8 are relatives. In how many ways can he invite 7 guests such that 5 of them are relatives?

→ Soln,

$$\text{Total friends} = 12$$

$$\text{Relatives} = 8$$

$$\text{Non-relatives} = 12 - 8 = 4$$

We have to select 7 friends out of which 5 must be relatives.

Relatives [8]	Non-Relatives [4]	Selection [7]
5	2	$C(8,5) \times C(4,2)$ $= 336$

$\therefore$ no. of selection = 336

d) There are 10 questions in group A of which 6 are to be solved. In group B there are only 6 questions of which 4 are to be solved. In how many ways an examinee can make up his choices?

→ Soln, Group-A

Total question = 10

Question to be solved = 6

Group-B

Total question = 6

Question to be solved = 4

Group A [10]	Group B [6]	Choices
6	4	$C(10,6) \times C(6,4)$ $= 3150$

$\therefore$ An examinee can make his choice in 3150 ways.

③

a) A bag contains 5 red balls. In how many ways at most 3 balls be selected?

↳ Soln,

Total balls = 5

We have to select at most 3 balls. That is, we may select ~~at~~ either 3 balls or 2 balls or 1 ball or 0 ball.

Ball [5]	selection
3	$C(5,3) = 10$
2	$C(5,2) = 10$
1	$C(5,1) = 5$
0	$C(5,0) = 1$

$$\therefore \text{Number of selection} = 10 + 10 + 5 + 1 \\ = 26 \quad \checkmark$$

b) Bidur has 4 coins of 1 rupee, 2 rupees, 5 rupees and 10 rupees each. How many different sums can be made by using these coins?

↳ Soln,

① ② ⑤ ⑩

To form different sums, we can select either 1 coin or 2 coins or 3 coins or all 4 coins.

coins [4]	selection
1	$C(4,1) = 4$
2	$C(4,2) = 6$
3	$C(4,3) = 4$
4	$C(4,4) = 1$

$$\therefore \text{no. of different sums} = 4 + 6 + 4 + 1 \\ = 15 \text{ ways}$$

b) There are 6 questions in a question paper. In how many ways can an examinee solve one or more questions.

↳ Soln,

d: Total questions = 6

Questions [6]	Solve
1	$C(6,1) = 6$
2	$C(6,2) = 15$
3	$C(6,3) = 20$
4	$C(6,4) = 15$
5	$C(6,5) = 6$
6	$C(6,6) = 1$

$$\text{No. of ways} = 6 + 15 + 20 + 15 + 6 + 1 \\ = 63 \text{ ways}$$

c) A candidate has to pass each of the five subjects to get through. In how many ways can the candidate fail?

↳ Soln,

Total subject = 5

Subject [5]	Ways to fail
1	$C(5,1) = 5$
2	$C(5,2) = 10$
3	$C(5,3) = 10$
4	$C(5,4) = 5$
5	$C(5,5) = 1$

$$\therefore \text{No. of ways} = 5 + 10 + 10 + 5 + 1 = 31$$

④ There are 15 cricket players in a club. In how many ways can a team of 11 be made? Also find the number of way if

- Two particular player are always included
- Two particular player are always included
- One particular player is always included and one particular is always excluded.

↳ Soln,

$$\text{No. of players} = 15$$

$$\text{No. of Team to be made} = 11$$

$$\begin{aligned} \text{No. of ways} &= C(15, 11) \\ &= 1365 \text{ ways} \end{aligned}$$

a) If two particular player are always included

$$\text{No. of total player} = 13$$

so,

$$\begin{aligned} \text{No. of ways} &= C(13, 11) \\ &= 78 \text{ ways} \end{aligned}$$

b) If two particular player are always included.

$$\text{No. of total player} = 15 \text{ but.}$$

$$\begin{aligned} \text{No. of total player to choose} &= 15 - 2 \\ &= 13 \end{aligned}$$

So,

$$\begin{aligned} \text{No. of ways} &= \text{Team to be made} = 11 - 2 \\ &= 9 \end{aligned}$$

Then,

$$\text{No. of ways} = C(13, 9)$$

$$= 715 \text{ ways}$$

c) One ~~particte~~ particular player is always included and one particular is always en-

cluded.

→ Soln,

No. of total player to choose = 13 (1 always excl
~~No. of team~~ to be cluded & 1 always included)

No. of total ~~player~~ to be made if 1 is
 always included = $11 - 1 = 10$

so,

$$\text{No. of ways} = C(13, 10) \\ = 286$$

5) In a paper of 2 groups having 5 questions each, in how many ways can 6 questions be answered if at least 2 questions from each group are to be attempted.

→ Soln,

Group A [5]	B [5]	sel [6]
2	4	$C(5, 2) \times C(5, 4)$
3	3	$C(5, 3) \times C(5, 3)$
4	2	$C(5, 4) \times C(5, 2)$

$$C(5, 2) \times C(5, 4) = 50$$

$$C(5, 3) \times C(5, 3) = 100$$

$$C(5, 4) \times C(5, 2) = 50$$

$$\therefore \text{No. of ways} = 50 + 100 + 50$$

$$= 200$$

✶

7) Out of 6 ladies and 8 gentlemen a committee of 11 is to be formed. In how many ways can this be done if the committee contains:

- Exactly 4 ladies.
- at least 4 ladies
- at most 4 ladies?

u) Soln,

Ladies = 6

Gents = 8

We have to select 11 members.

① Ladies [6]	Gents [8]	Selection [11]
4	7	$C(6,4) \times C(8,7) = 120$

$\therefore$ No. of selection = 120 u.

② Ladies [6]	Gents [8]	Selection [11]
4	7	$C(6,4) \times C(8,7) = 120$
5	6	$C(6,5) \times C(8,6) = 168$
6	5	$C(6,6) \times C(8,5) = 56$

$\therefore$ No. of selection = $120 + 168 + 56$
 $= 344$ u.

③ Ladies [6]	Gents [8]	Selection [11]
4	7	$C(6,4) \times C(8,7) = 120$
3	8	$C(6,3) \times C(8,8) = 20$

$\therefore$ No. of selection = $120 + 20$
 $= 140$ u.

⑥ Out of 5 members belonging to party A and 6 members belonging to party B, in how many ways can a committee of 8 members be formed so that party B has always majority in committee?

→ Soln,

For party B has always majority in committee.

Party A [5]	party B [6]	Set [8]
2	6	$C(5,2) \times C(6,6) = 10$
3	5	$C(5,3) \times C(6,5) = 60$

$$\therefore \text{No. of ways} = 10 + 60 \\ = 70$$